TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2018

CANDIDATE NAME							
CENTRE NUMBER	S						
INDEX NUMBER							
CLASS	C	G			/	1	7

H1 BIOLOGY

Multiple Choice

8876/01

Wednesday 14 September 2018

1 hour

Additional materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice in **soft pencil** on the separate Multiple Choice Answer Sheet.

Read the instructions on the Multiple Choice Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **18** printed pages.

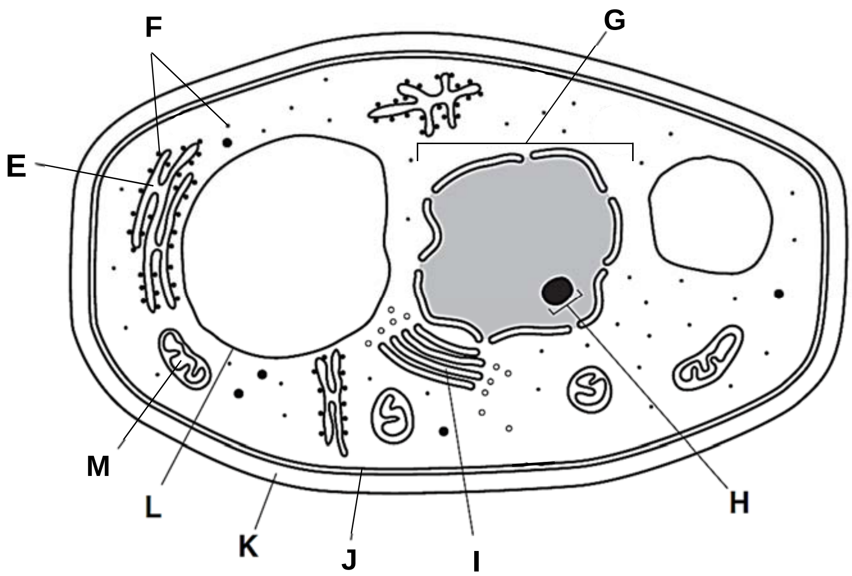
Section A

Answer **all** the questions in this section.

1 Tuberculosis and candidiasis are two opportunistic infections that may develop during AIDS.

Tuberculosis is caused by *Mycobacterium tuberculosis*, a prokaryote that lives in human lungs; whereas candidiasis is caused by *Candida albicans*, a yeast-like fungus that lives in human lungs.

The figure below shows the structure of *Candida*.



Which of the structure(s) can also be found in the causative agent that causes tuberculosis?

A None

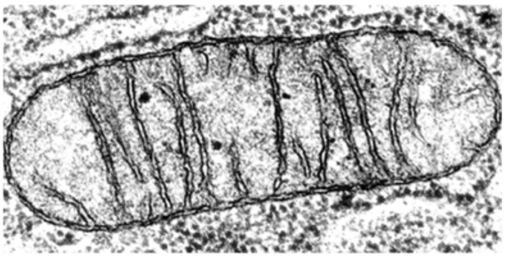
B **F** only

C **F**, **J**, **K** only

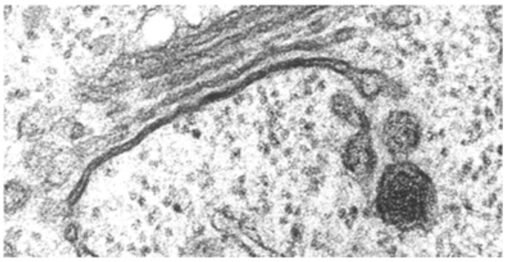
D **H**, **J**, **K** only

2 The images below show the electron micrographs of some organelles found in eukaryotic cells.

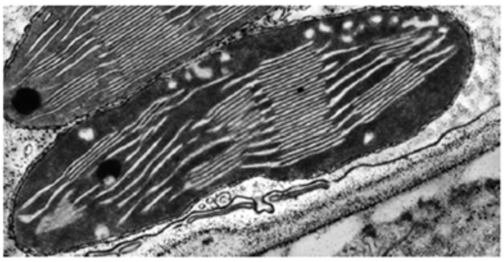
P



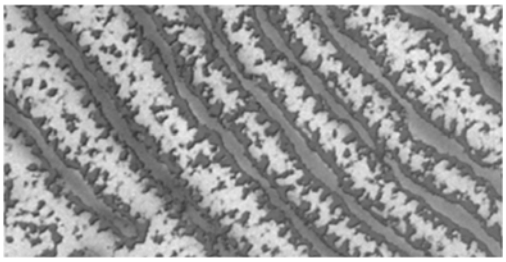
Q



R



S



The following statements are descriptions of membranous cell structures.

1 formed by a single membrane and enclosing a large fluid-filled space and regulating the osmotic pressure of the cell

2 formed by a single membrane and enclosing inactivated enzymes

3 formed by a single membrane that has flattened sacs and tubular structures inter-connected throughout the cell, sometimes with a complex of nucleic acid and protein attached

4 formed by a single membrane that has tubular structures and containing enzymes to add carbohydrate side chains to proteins

5 formed by two membranes and internal membranes that contain pigments

6 formed by two membranes whereby the inner membrane is folded extensively

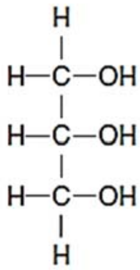
7 formed by two membranes, the outer membrane is continuous with another membranous organelle

Which of the following row correctly matches the descriptions of the cell structures?

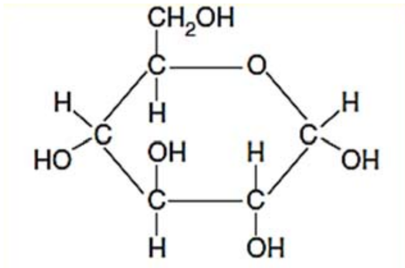
	P	Q	R	S
A	5	3	6	1
B	5	2	4	7
C	6	4	5	3
D	7	1	2	6

3 Which molecule is found in glycogen?

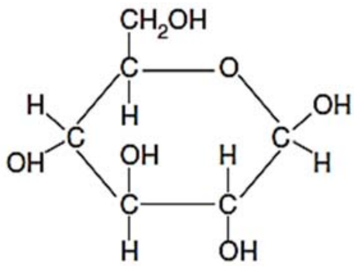
A



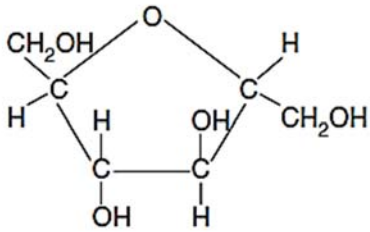
B



C



D



4 Particular biological molecules react with chemicals called reagents to give distinct colour changes. The colour depends on the kind of biological molecule and the type of reagent used, as shown in the following table.

chemical reagent	biological molecule	colour change observed
L	protein	violet
M	lipid	red
N	nucleic acid	green

A researcher added different reagents to some isolated ribosomes.

The colour change observed are

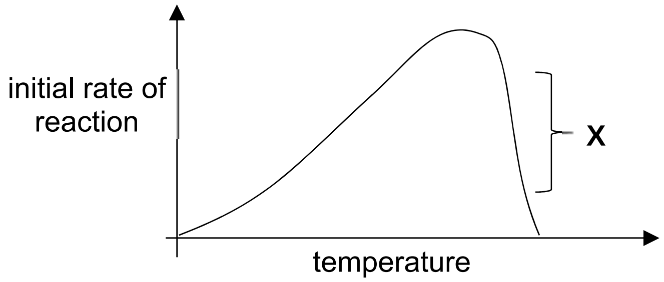
A green only.

B red and green.

C green and violet.

D violet, red and green.

5 The diagram shows the initial rate of reaction using constant amounts of substrate and enzyme at different temperatures.



What is the reason for the decline in the level of activity in region **X**?

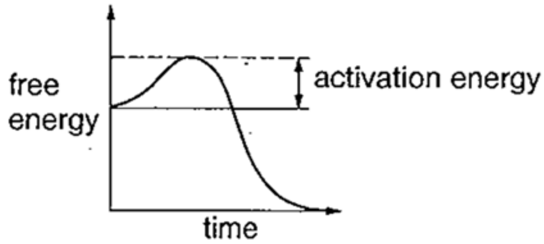
A breaking of sulphur bridges and ionic bonds in the enzymes

B competition between substrate and product for the active site

C breaking of hydrogen bonds and hydrolysis of peptide bonds in the enzyme

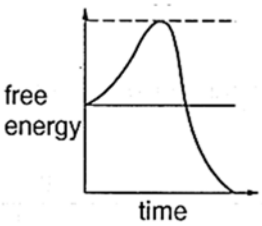
D insufficient substrates to occupy all the active sites

6 The graph shows energy changes during an uncatalysed chemical reaction.

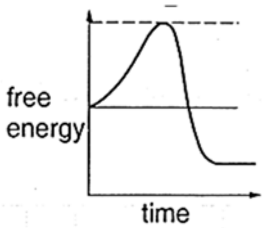


Which graph shows the energy changes for the same reaction when it is catalysed by an enzyme?

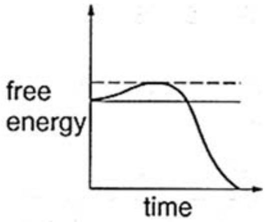
A



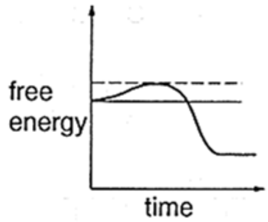
B



C



D



7 Some of the molecules found in animal tissues are grouped into three lists.

1 glucose, cholesterol, triglycerides, water

2 glycogen, adenine, phospholipids

3 haemoglobin, carbon dioxide, mRNA, fructose

Which lists include one or more molecules that always contain nitrogen atoms?

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

8 Proteins in the cell surface membranes of human cells and mouse cells were labelled with red and green fluorescent dyes respectively.

When a human cell and a mouse cell were fused together, the red and green fluorescent dyes were at first found in different regions of the cell surface membrane of the hybrid cell, but after 40 minutes, they were evenly distributed in the entire cell surface membrane.

What explains this observation?

A All protein molecules in the cell surface membrane are fixed to structures within the cell, but phospholipid molecules move freely between them.

B Groups of protein and phospholipid molecules in the cell surface membrane are attached to each another and move together.

C Only protein molecules in the outer layer of the cell surface membrane can move freely between phospholipid molecules.

D Protein molecules in the outer layer of the cell surface membrane and those which span the bilayer can move freely between phospholipid molecules.

9 At prophase of mitosis, a eukaryote chromosome consists of two chromatids.

What is the structure of a single chromatid?

A one molecule of single-stranded DNA coiled around protein molecules

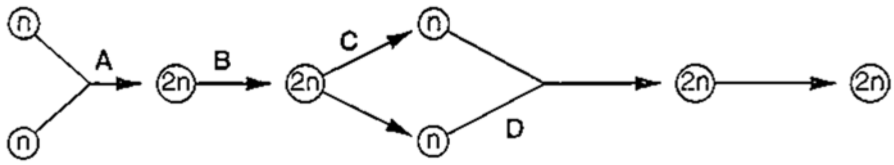
B two molecules of single-stranded DNA each coiled around protein molecules

C one double helix of DNA coiled around protein molecules

D two double helices of DNA each coiled around protein molecules

10 The diagram represents the life cycle of an animal.

At which stage in the life cycle does mitosis occur?



11 Which is the correct statement concerning cell and nuclear division?

A Haploid eukaryotes can reproduce by mitosis whereas diploid eukaryotes can reproduce by mitosis or meiosis.

B Just before prophase, the mass of DNA is double the normal mass. Following anaphase, this mass is reduced by half and following cytokinesis this mass halves again.

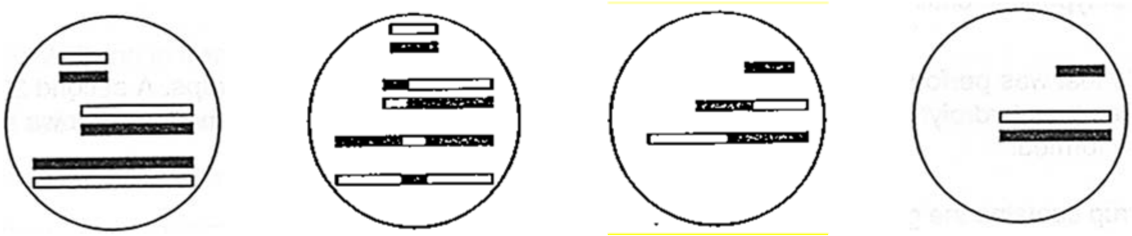
C Mutagens can cause mutations whereas carcinogens can cause cancer. This means that all mutagens are carcinogenic.

D Some of the roles of mitosis are growth, asexual reproduction, cell repair following tissue damage and cell replacement.

12 The diagram shows the maternal and paternal chromosomes from a diploid cell.

If the cell divides by meiosis, which diagram shows a possible viable gamete?

A **B** **C** **D**



13 Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine and the blood.

However, within their own environments, a blood cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a blood cell.

Which statement explains this?

A Different stem cells have only the genes required for their particular cell line.

B Genes not required for the differentiation of a particular cell line are methylated.

C Binding of repressor molecules prevents the expression of genes not required for a particular cell line.

D Expression of gene not required for a particular cell line is controlled at translational level.

14 Which row represents the correct features of the nitrogenous base adenine?

	has a single ring structure	is a purine	joins its complementary base by three hydrogen bonds	pairs with uracil in RNA
A	✓	✓	✗	✗
B	✓	✗	✓	✗
C	✗	✓	✗	✓
D	✗	✗	✗	✓

key

✓ = true

✗ = false

15 The table shows the mode of action of two antibacterial drugs that can affect the synthesis of proteins.

antibacterial drug	rifampicin	streptomycin
mode of action	binds to RNA polymerase	causes errors in translation

If bacteria are treated with both drugs, what will be the immediate effects?

1 Transcription will stop, but non-functional proteins may continue to be synthesised.

2 If translation has started, proteins may be non-functional.

3 Translation will be inhibited.

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

16 A peptide consists of ten amino acids of four different kinds.

What is the theoretical minimum number of different kinds of tRNA molecules required to translate the mRNA for this peptide?

A 4

B 10

C 12

D 30

17 Activation of an amino acid for translation requires

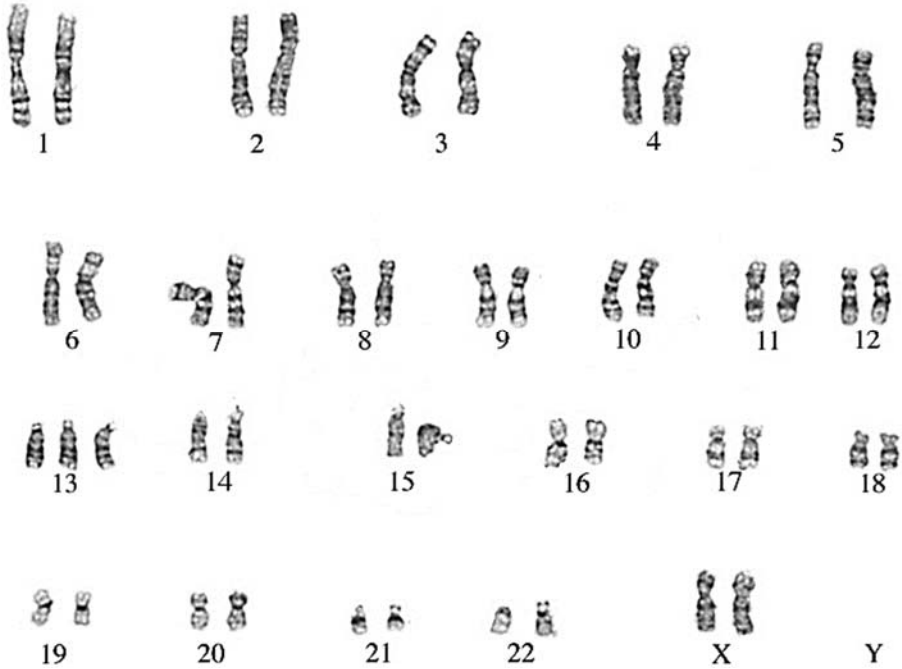
A joining the amino group of the amino acid to the 5'-end of the tRNA.

B matching the anticodon of the tRNA to the codon of the mRNA.

C hydrolysis of ATP.

D one enzyme that is specific for both a particular amino acid and a particular tRNA.

18 A newborn baby was diagnosed with Patau syndrome. The diagram below shows her chromosomes.



This is an example of

A frameshift mutation

B silent mutation

C aneuploidy

D polyploidy

19 Cancer cells divide out of control, forming tumours.

Which statement describes the difference between a cancer cell and a normal cell?

A Cancer cells do not undergo cytokinesis

B Cancer cells have a shorter interphase

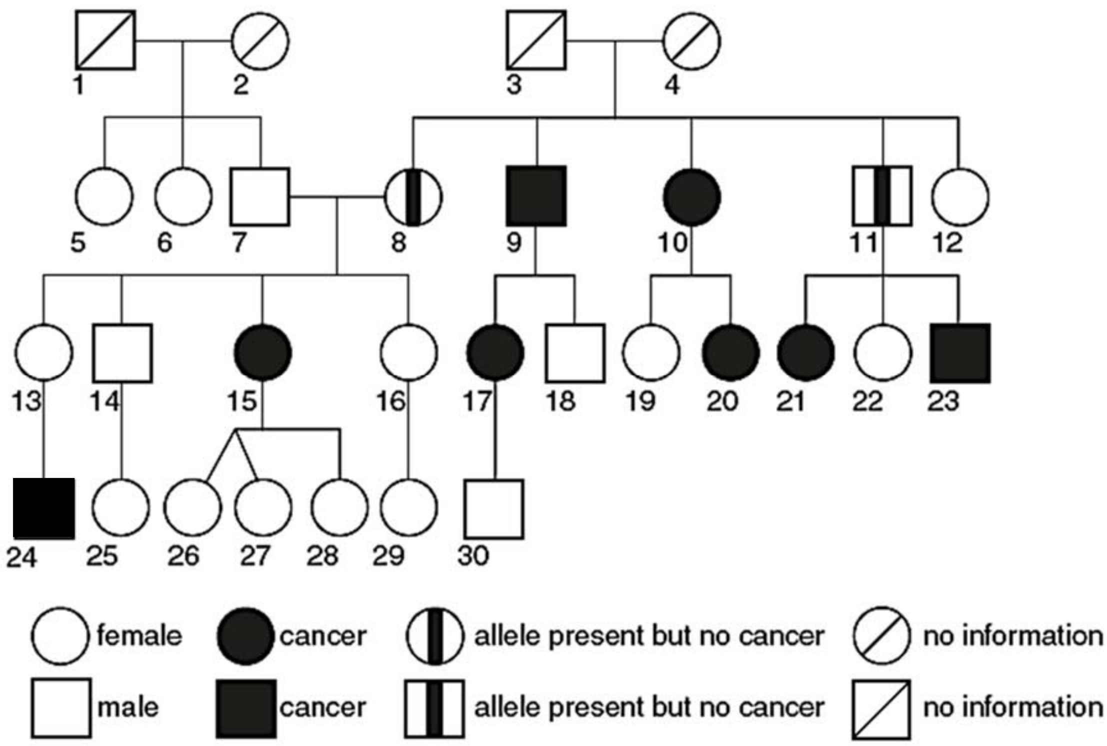
C Cancer cells do not have metaphase

D Only cancer cells have mutated DNA

20 The BRCA2 protein is involved in suppressing the development of tumours. The gene that codes for this protein is on chromosome 13.

Several different dominant alleles of this gene, *BRCA2*, code for faulty versions of the protein. The presence of any one of these faulty alleles leads to an increased chance of developing several types of cancer, including breast cancer. Not everyone with one of these alleles develops cancer.

The pedigree (family tree) below shows the occurrence of cancers in four generations of a family. The presence of a faulty *BRCA2* allele was confirmed in person 15. The other individuals with cancer were not tested for the presence of the allele. For individuals 17 to 30, only one of their parents is shown in the pedigree. Individuals 24–30 are all under twelve years old.



Which one of the following statement is **not** correct?

A Individuals 8 and 11 have *BRCA2* allele and may develop cancer later in life.

B Individuals 8 to 11 may have inherited *BRCA2* allele from either of their parents.

C Individual 15 may have inherited one copy of *BRCA2* allele from her mother.

D Individual 24 may have inherited the *BRCA2* allele only from his mother and not his father.

21 In mice, the gene for 'dappled' coat (**D**) and its recessive allele, the gene for 'plain' coat (**d**), are located on the X chromosome. The gene for 'straight' whiskers (**W**) and its recessive allele, the gene for 'bent' whiskers (**w**), are autosomal.

A male mouse with plain coat and bent whiskers was mated on several occasions to the same female and the large number of offspring consisted of the following phenotypes in equal proportion:

dappled male with straight whiskers
dappled female with straight whiskers
dappled male with bent whiskers
dappled female with bent whiskers

plain male with straight whiskers
plain female with straight whiskers
plain male with bent whiskers
plain female with bent whiskers

If $\mathbf{X}^D$ represents an **X** chromosome carrying an allele for 'dappled' coat and $\mathbf{X}^d$ represents an **X** chromosome carrying an allele for 'plain' coat, what is the genotype of the female parent?

A $\mathbf{X^DX^DWW}$

B $\mathbf{X^DX^DWw}$

C $\mathbf{X^DX^dWW}$

D $\mathbf{X^DX^dWw}$

22 A trial breeding programme between Nepalese yaks and a breed of British cattle called the Dexter was carried out to develop a hybrid that was hardy, easy to handle, produced good quality meat and high milk yield. The preliminary results of the trial showed the relative strengths of the alleles of the genes for desired characteristics.

desired characteristic	Dexter	Yak	hybrid
aggression	low	high	low
intelligence	low	high	high
hardiness	low	high	high
meat quality	high	low	high
milk yield	high	low	high

Which combination shows the animals that appear to have alleles dominant for the desired characteristic?

	aggression	intelligence	hardiness	meat quality	milk yield
A	Dexter	Dexter	Dexter	Yak	Yak
B	Dexter	Yak	Yak	Dexter	Dexter
C	Yak	Dexter	Dexter	Dexter	Dexter
D	Yak	Dexter	Dexter	Dexter	Yak

23 *Staphylococcus aureus* is a common bacterium, found on human skin. There are many strains of *S. aureus*. The antibiotic methicillin was then used to treat infection by *S. aureus*. Now there are at least 15 different strains of MRSA (methicillin resistant *Staphylococcus aureus*).

Which of the following are valid reasons for the emergence of 15 different strains of MRSA?

1 The bacteria mutated when it was exposed to methicillin, thus becoming resistant.

2 The bacteria underwent spontaneous mutation and some strains happened to be resistant to methicillin.

3 The antibiotic caused the bacteria to produce methicillin-resistant proteins.

A 2 only

B 1 and 2

C 2 and 3

D 1, 2 and 3

24 During aerobic respiration ATP can be formed by glycolysis and oxidative phosphorylation in the electron transport system.

In the complete oxidation of one molecule of glucose, approximately what percentage of ATP is formed by oxidative phosphorylation?

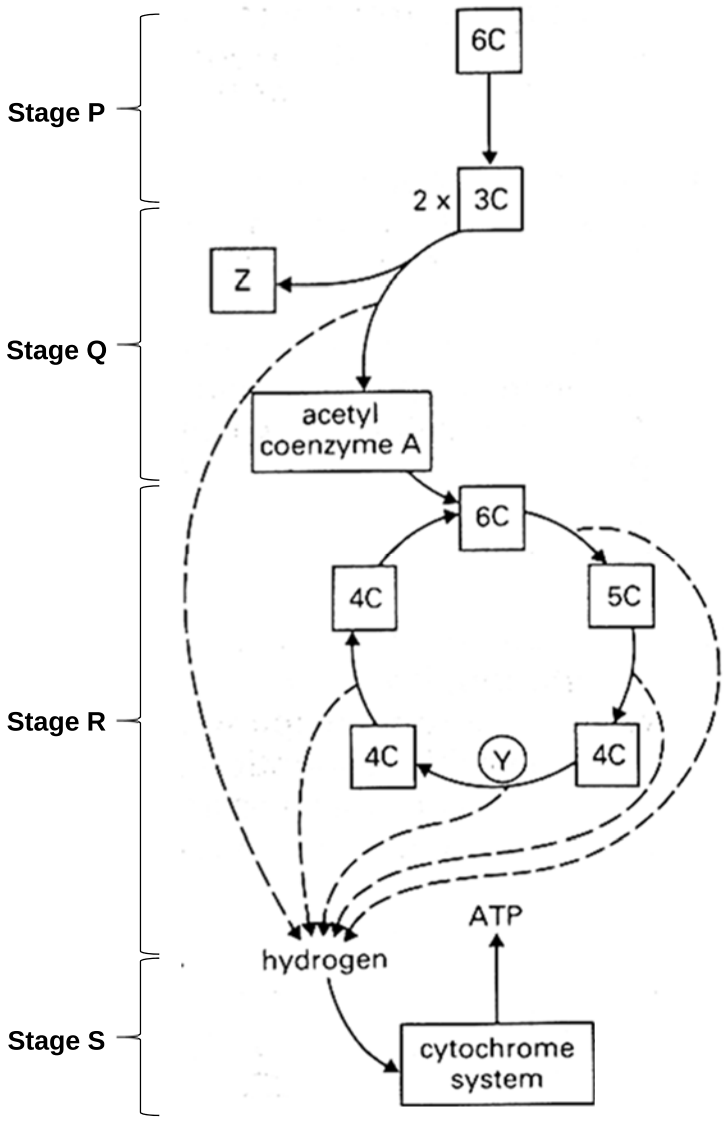
A 10%

B 25%

C 75%

D 90%

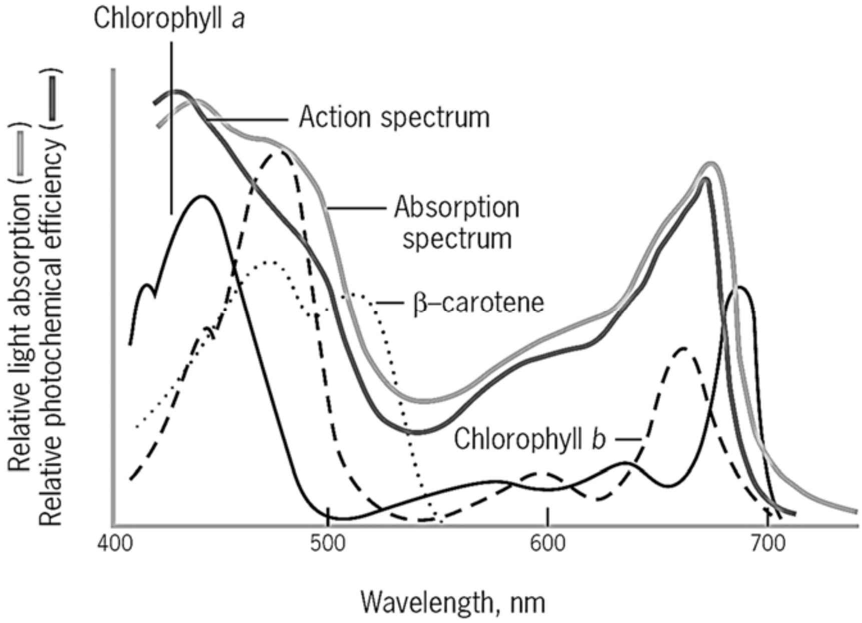
25 The diagram shows a biochemical pathway in a typical eukaryotic cell.



Which of the following row is correct?

	cristae	cytosol	mitochondrial matrix	final electron acceptor
A	**R**	**P, Q**	**R**	NADP
B	**R**	**P, Q**	**R**	Oxygen
C	**S**	**P**	**Q, R**	Oxygen
D	**S**	**P**	**Q, R**	NADP

26 The figure below shows the absorption spectrum of the photosynthetic pigments of a flowering plant and its action spectrum.



What can be concluded from the graph above?

1 The relative light absorption will be higher at higher temperatures, as temperature is a limiting factor.

2 The green leaves reflect light of wavelength 550 nm, hence the photochemical efficiency is low.

3 The compensation point of β-carotene, whereby the rate of photosynthesis equals the rate of respiration, occurs at 550nm.

4 The accessory pigments chlorophyll b and β-carotene absorb light energy mostly at 480nm.

A 2 and 4

B 1, 2 and 3

C 1, 3 and 4

D All of the above

27 Homogenized leaf suspensions containing the cytoplasm and organelles were then placed in two different test-tubes.

Test-tube **A** contains non-labelled water ($H_2{}^{16}O$)

Test-tube **B** contains radioactively-labelled water ($H_2{}^{18}O$)

A few drops of DCPIP, a hydrogen acceptor, were added to each test-tube. DCPIP will turn from blue to colourless when it is reduced and this colourless DCPIP can be reoxidized to blue.

The test-tubes were then exposed to red light for 30 minutes.

Which of the following shows the results of the two test-tubes after 30 minutes?

	Tube **A**		Tube **B**	
	Gas evolved	DCPIP colour	Gas evolved	DCPIP colour
A	$^{16}CO_2$	Blue	$^{18}CO_2$	Blue
B	$^{16}CO_2$	Colourless	$^{18}CO_2$	Colourless
C	$^{16}O_2$	Blue	$^{16}O_2$	Blue
D	$^{16}O_2$	Colourless	$^{18}O_2$	Colourless

28 An investigation was carried out to assess the effect of diet on the milk yield and methane production of cows. The cows in group **A** were fed a traditional diet and those in group **B** were fed the same diet with a mixture of chopped hay and straw added.

The table below shows the results of this investigation.

Group	Diet	Mean milk yield per cow/ $dm^3\ day^{-1}$	Methane emission for each dm^3 milk produced / dm^3
A	Traditional with no added material	24.0	30.0
B	Traditional with added chopped hay and straw	27.6	24.0

Which of the following actions will help reduce the impact of global warming?

1 Decreasing consumption of beef and milk

2 Creating more foraging grounds to feed the cows

3 Adding chopped hay and straw to the cows' diet.

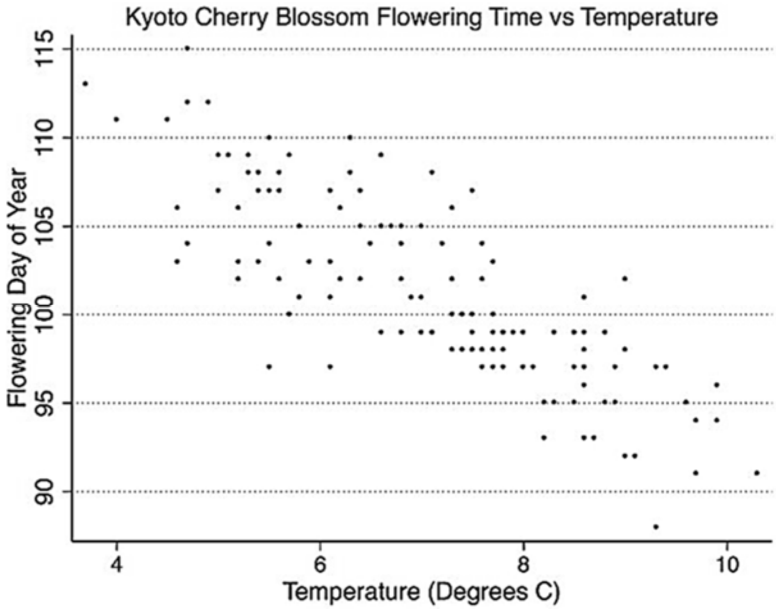
A 1 only

B 1 and 3 only

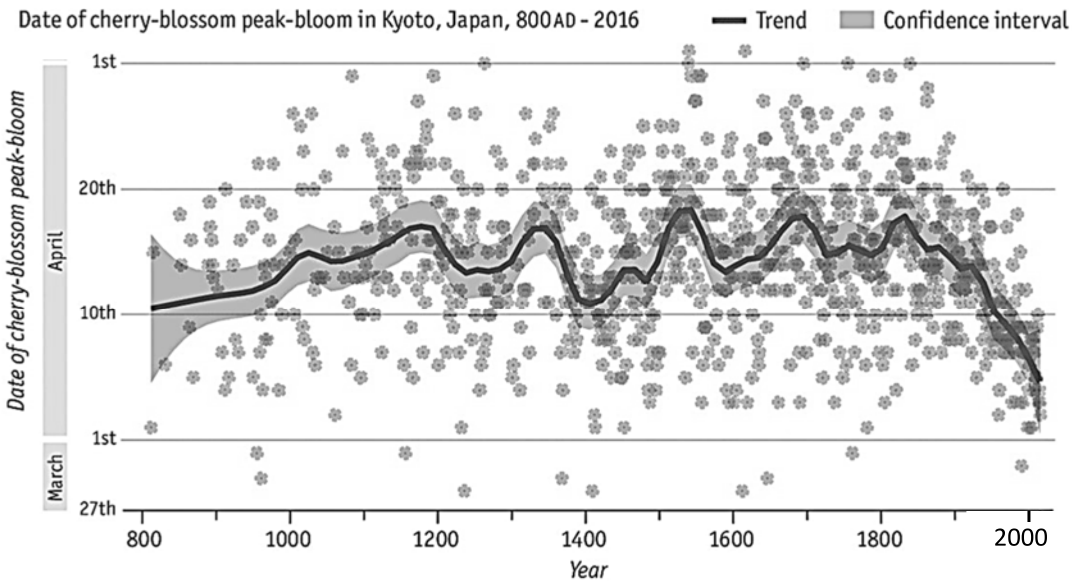
C 2 and 3 only

D All of the above

29 The effect of temperature on cherry blossom flowering time (day of the year) is shown below.



The records of timing of cherry blossoms in Japan from 800 A.D is shown below.



Which conclusions can be made from both graphs?

1 The peak of cherry blossoms has consistently been earlier since 1850. Cherry blossoms begin earlier as temperature increases from 4 to 10°C.

2 The temperature in Japan has been increasing since 800 A.D., resulting in later blooming and pollinators are unable to pollinate the cherry trees.

3 No conclusion can be made as the data points are scattered and lack clear trend.

A 1 only

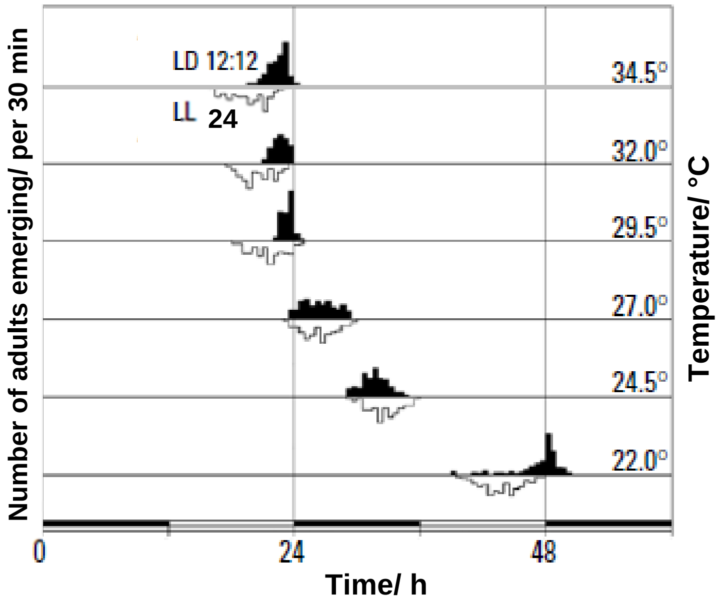
B 2 only

C 3 only

D 1 and 2

30 The figure below shows the effect of temperature on the emergence of a particular species of mosquitoes.

The mosquitoes were bred either in a constant 24 hours light condition (LL white bars), or a 12 hours light followed by 12 hours dark condition (LD 12:12 black bars).



Which of the following row is correct?

	life cycle of mosquito	descriptions
A	egg → larva → pupa → adult	As temperature increases, the timing in which the adult mosquitoes emerged increases from 24h to 48h
B	egg → larva → pupa → adult	As temperature increases, the timing in which the adult mosquitoes emerged decreases from 48h to 24h
C	egg → pupa → larva → adult	As temperature increases, the timing in which the adult mosquitoes emerged increases from 24h to 48h
D	egg → pupa → larva → adult	As temperature increases, the timing in which the adult mosquitoes emerged decreases from 48h to 24h